

Lecture 13 - Background for Nyquist

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13.1 Overview

Today:

- Recap stability margins
- Complex analysis
- Cauchy's Argument Principle

Next week:

- Nyquist plot
- Nyquist stability & special cases
- Examples

13.1.1 Recap: : Stability Margins

We care about gain and phase margin because they let us quantify how safe we are.

- How much gain and phase we can tolerate before making the system unstable.
- Many tools in classical control theory help us evaluate stability in different ways, including Routh Hurwitz, gain and phase margin using Bode, and today: Nyquist Plots.
- Routh Hurwitz gives us bounds on K for stability, and using stability margins on Bode plots gives us a graphical way to easily meet design goals.
- Because of uncertainties in our model, control engineers can give design requirements specifying a minimum gain margin or phase margin.
- Example: Choose K to give a gain margin of 20 dB

$$\underbrace{40.8}_{\text{old GM}} - \underbrace{20}_{\text{New GM}} = 20.8 \text{ dB}$$
$$20\log(K) = 20.8$$
$$K = 10^{1.04} = 11$$

However:

- If multiple crossovers then it is hard to evaluate. And more importantly:

- Bode margins visually describe gain and phase margin, but only defined for stable systems with LHP-zeros where increasing gain K makes the system go unstable.

→ Solution: Nyquist.

Recall:

- We need $G_{RY} = \frac{GH}{1+GH}$ to be stable
- G_{RY} is unstable, i.e. $G_{RY} \rightarrow \infty$, if:

$$1 + GH = 0$$

- This occurs if $GH = -1$, i.e. when:

$$\text{Magnitude } |GH| = 1$$

$$\text{Phase } \angle GH = -180^\circ$$

13.1.2 Gain & Phase From Zeros & Poles

Show values associate Gain and Phase of combined zeros and poles $\frac{\text{Mag Z}}{\text{Mag P}}$.

Example: stable Pole at $s = -1$:

$$G(s) = \frac{1}{s+1}$$

Mag:

$$\begin{aligned} G(i\omega) &= \frac{1}{i\omega+1} \cdot \frac{1-\omega i}{1-\omega i} \quad s = i\omega \text{ response to sinusoid of freq } \omega \\ &= \frac{1-i\omega}{1^2-\omega^2 i^2} = \frac{1-i\omega}{1+\omega^2} \quad \text{Will use this last fraction for phase too} \\ |G(i\omega)| &= \frac{\sqrt{1^2+\omega^2}}{1+\omega^2} = \frac{1}{\sqrt{1+\omega^2}}, \quad \omega > 0 \end{aligned}$$

Phase:

$$\begin{aligned} \text{Re}(G(i\omega)) &= \frac{1}{1+\omega^2}, \quad \text{Im}(G(i\omega)) = \frac{-\omega}{1+\omega^2} \\ \phi &= \tan^{-1}\left(\frac{-\omega}{1}\right), \quad \omega > 0 \end{aligned}$$

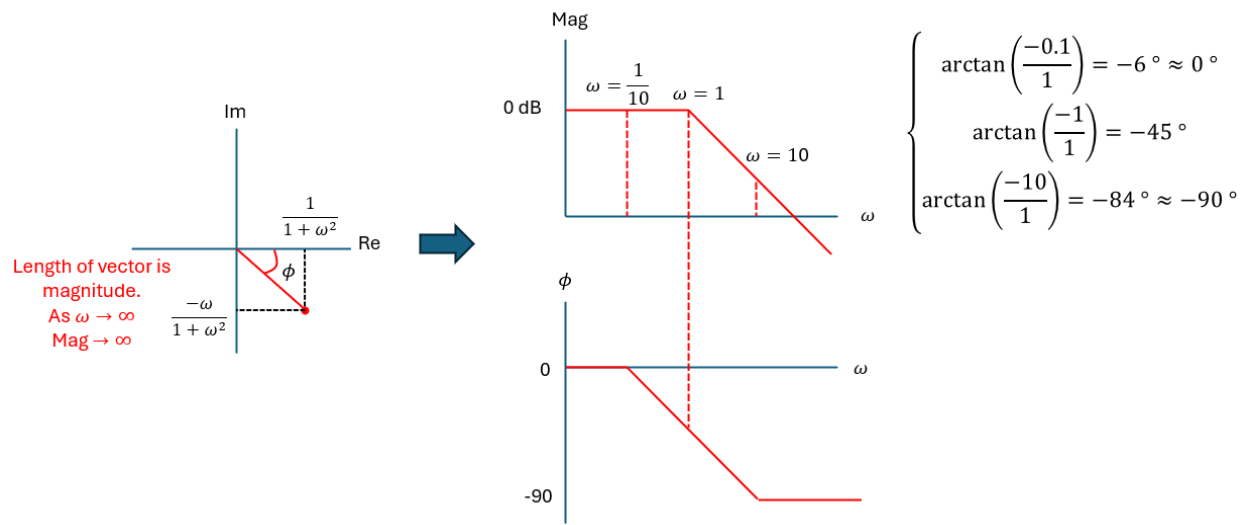


Figure 13.1: Breakdown of how Bode plots get generated.

Example: Pole at $s = 0$

This example helps explain why we call $\frac{1}{s}$ an "integrator".

$$G(s) = \frac{1}{s} \rightarrow G(i\omega) = \frac{1}{\omega i} = \frac{i}{\omega i^2} = \frac{1}{-\omega} i$$

$$\Rightarrow \text{Mag: } |G| = \left| \frac{1}{\omega} \right|$$

$$\Rightarrow \text{Phase: } \arg(G) = \arg\left(-\frac{1}{\omega}, 0\right) = -90^\circ$$

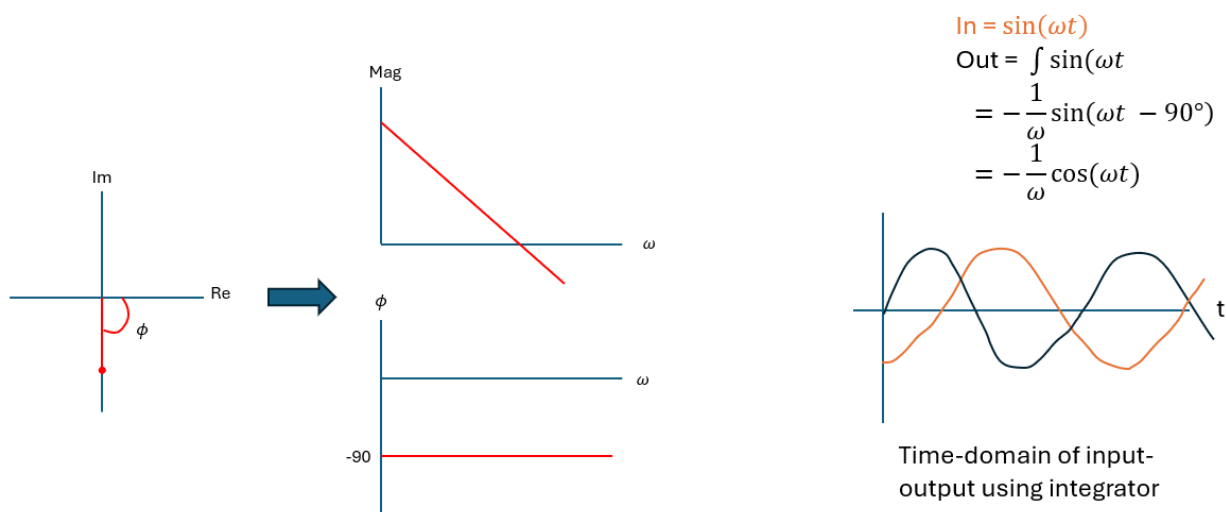


Figure 13.2: Integrator visually.

Example: Unstable Pole at $s = 1$

$$\begin{aligned}
 G(s) &= \frac{1}{s-1} \\
 G(i\omega) &= \frac{1}{i\omega-1} \cdot \frac{-i\omega-1}{-i\omega-1} \\
 &= \frac{-1-i\omega}{1+\omega^2} \implies \text{Re: } \frac{-1}{1+\omega^2}, \quad \text{Im: } \frac{-\omega}{1+\omega^2} \\
 |G(i\omega)| &= \frac{\sqrt{1^2+\omega^2}}{1+\omega^2} \\
 &= \frac{1}{\sqrt{1+\omega^2}}
 \end{aligned}$$

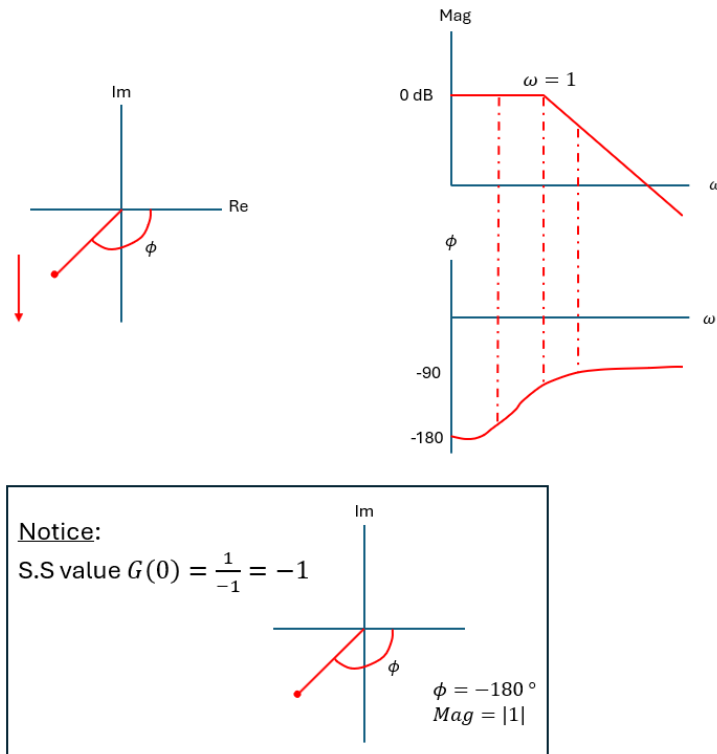


Figure 13.3: Unstable pole example.

13.1.3 Multiple Poles and Zeros

$$G(s) = \frac{(s-z_1)(s-z_2)\dots}{(s-p_1)(s-p_2)\dots}$$

Let $Z_i = s - z_i$ and $P_i = s - p_i$

$$\text{Combined Magnitude: } |G(s)| = \frac{|Z_1||Z_2|\dots|Z_N|}{|P_1||P_2|\dots|P_N|} \Rightarrow \frac{\prod |Z_i|}{\prod |P_i|}$$

Know this from Bode, where we multiply gains by adding them on log plot.

$$\text{Combined Phase: } \angle G(s) = \angle Z_1 + \angle Z_2 + \dots + \angle Z_N - \angle P_1 - \angle P_2 - \dots - \angle P_N \Rightarrow \sum \angle Z_i - \sum \angle P_i$$

Why? $\rightarrow Z_i = s - z_i = iw - z_i = |a|e^{i\theta}$
 $Ae^{i\theta_1} * Be^{i\theta_2} = AB e^{i(\theta_1+\theta_2)}$

Just like Bode, e.g. 1 pole: $0 \rightarrow -90^\circ$, 2 poles: $0 \rightarrow -180^\circ$

13.1.4 Cauchy's Argument Principle

Relates the number of poles and zeros inside a contour in one complex plane to the characteristics of a plot in another complex plane.

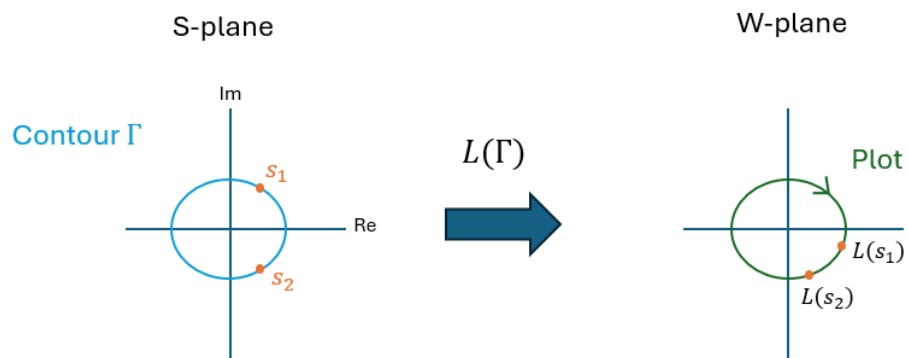


Figure 13.4: Cauchy's Argument Principle.

- We define a "contour", i.e. a closed curve in the S-plane.

$\rightarrow$ This contour is just an ordered sequence of points in the S-plane (s_1 , s_2 , etc).

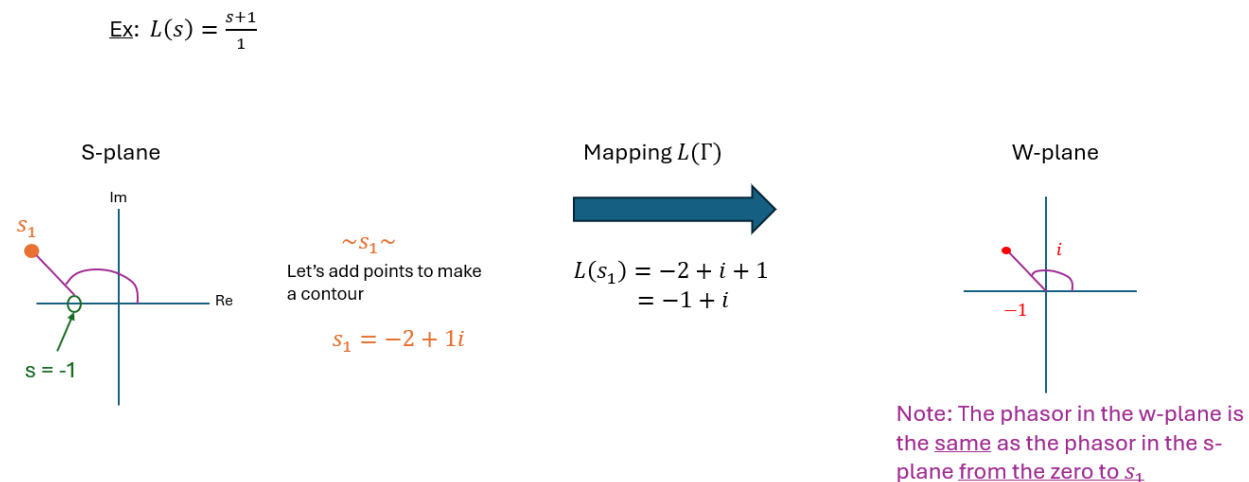
- L is a complex-valued function - in our case, a transfer function - which maps these points (complex numbers) to a new set of complex numbers in the W-plane.

$$\underbrace{\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\}}_{\text{Set of complex numbers}}, \quad L : \mathbb{C} \rightarrow \mathbb{C}$$

Where L takes complex numbers as inputs and outputs complex numbers.

Let us try to understand this better,

- L is a transfer function, and we can understand the operation $L(\Gamma)$ by showing L on the S-plane,

Figure 13.5: Example of the mapping $L(\Gamma)$.

- Let's pick 3 more points and complete our contour by returning to s_1 ,

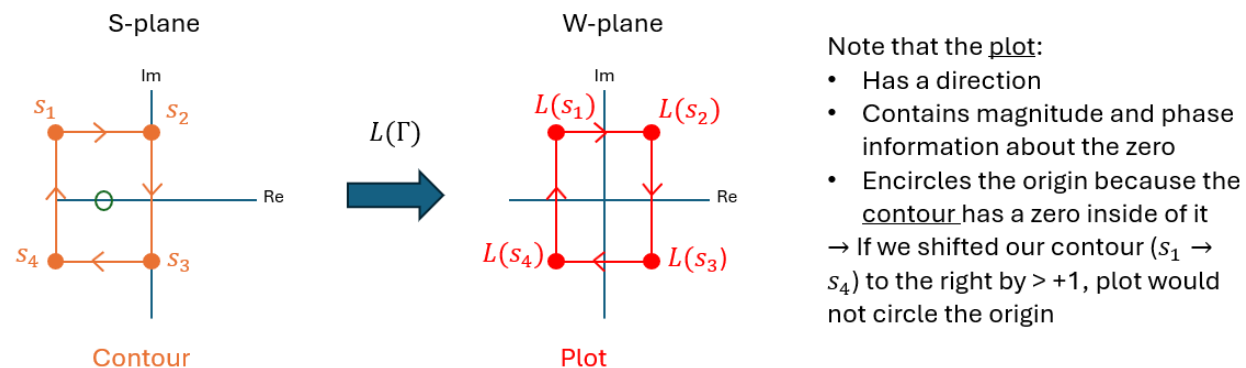


Figure 13.6: Completing the contour and mapping.